ISSN 1028-334X, Doklady Earth Sciences, 2014, Vol. 456, Part 1, pp. 512–516. © Pleiades Publishing, Ltd., 2014.
Original Russian Text © N.M. Fedorova, N.M. Levashova, J.G. Meert, A.V. Maslov, M.T. Krupenin, 2014, published in Doklady Akademii Nauk, 2014, Vol. 456, No. 1, pp. 77–81.

GEOLOGY

New Paleomagnetic Data on Baltica Based on Upper Ediacaran Deposits on the Western Slope of the Middle Urals

N. M. Fedorova, N. M. Levashova, J. G. Meert,
Corresponding Member of the RAS A. V. Maslov, and M. T. Krupenin

Received November 26, 2013

Abstract—The latest concepts about the Earth's paleogeography for the period of 700–500 million years are quite contradictory. Reliable paleomagnetic data are quite scarce for the Ediacaran–Cambrian of the majority of continental plates, which means that making any reliable global paleogeographic and paleotectonic reconstructions is impossible. According to various authors, Baltica within this time, for example, could have been located at any latitudes from the South Pole to the equator. Making correct reconstructions requires new paleomagnetic data; however, almost all objects that are applicable for such studies within Baltica have already been studied. A possible solution is to study the deformed margins of the plate, in particular, the western megazone of the Middle Urals, where the lower and upper Ediacaran volcanogenic–sedimentary and sedimentary sections are known within the Kvarkushsko–Kamennogorskii anticlinorium [1, 2] (Fig. 1). This paper presents the first paleomagnetic results obtained for the sedimentary rocks of the Upper Ediacaran Chernokamenskaya suite. They are consistent with the group of six poles of the same age [3–7], by which Baltica was located at the subequatorial latitudes at the end of the Educarian.

DOI: 10.1134/S1028334X14050134

The Chernokamenskaya suite 1500–1800 m thick is an alternation of greenish purple–gray, cherry-blossom–red, and reddish–brown fine-grained polymictic sandstones, siltstones, and argillites [1]. In the lower section of the formation, there are interlayers of volcanic tuffs, whose zircon isotopic U–Pb age is 557 ± 13 million years [8].

Paleomagnetic sampling of rocks from the Chernokamenskaya suite was made in ten natural outcrops in the valleys of the Vil'va and Vizhai rivers (Fig. 1B). The samples were selected manually out of the reddish–brown, cherry-blossom–red, and purple–gray siltstones and fine-layered silty sandstones. The samples were oriented by a mining compass. The thickness of the sampled intervals ranged from 2–3 to 15–20 m. The sampling was done in stratigraphic order by the sites (7–8 samples in each). In total 39 sites were analyzed (approximately 270 samples).

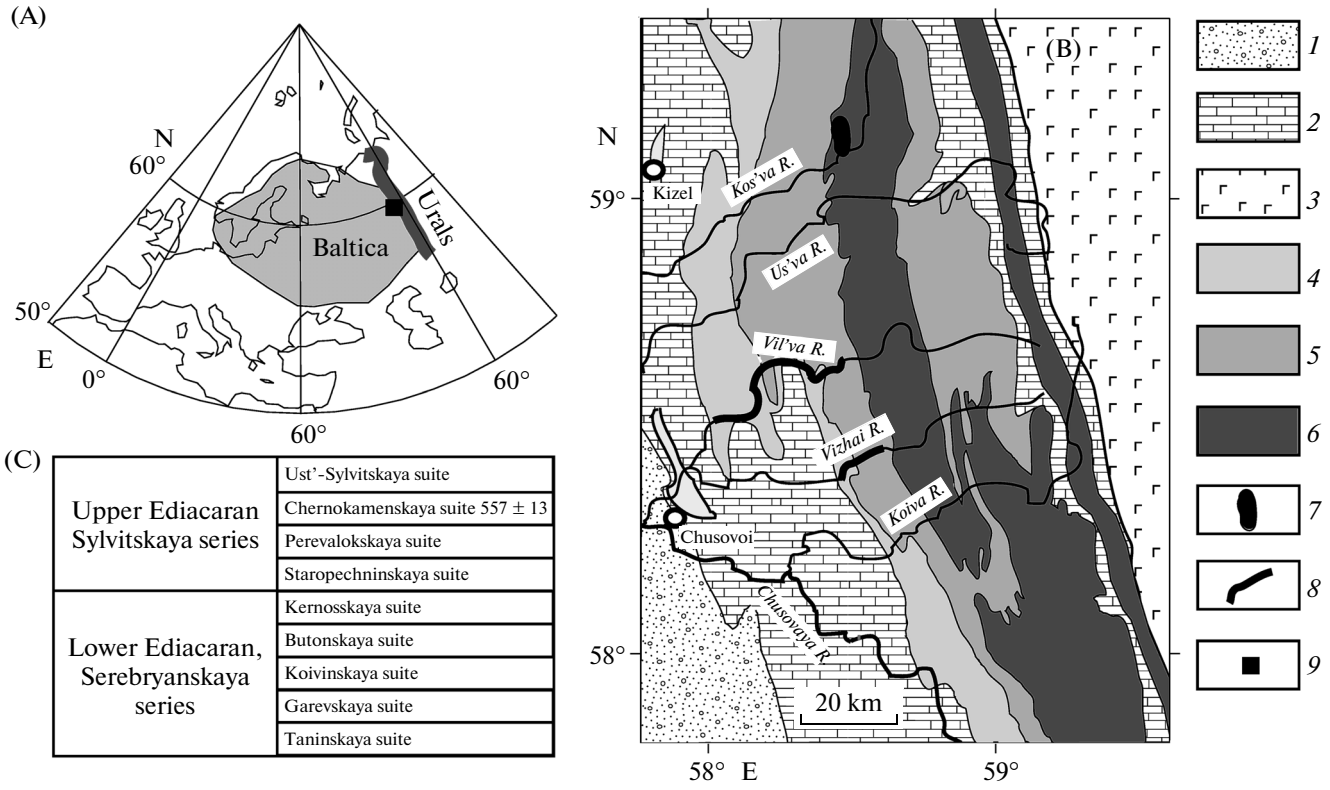
Cubes with an edge of 20 mm were cut from the samples and then were subject to stepwise (up to 20 steps) thermal demagnetization to 700°C. The devices and procedure for the demagnetization corresponded to [3]. The data obtained were analyzed using orthogonal diagrams. The directions of the magnetization components were determined by the linear sections of the trajectories that included at least three measurements [9]. The total average trend for the entire stratum was calculated by the average trends for the sites.

The majority of the samples exhibited an unstable, poorly grouped magnetization component at temperatures up to 200°C. It does not have physical significance and will not be considered further.

When heated at above 200°C, approximately 60% of the samples produce 1–2 magnetization components. The average-temperature component (ATC) is detected in a range of 200–480°C (Figs. 2A, 2C), and the high-temperature component (HTC), in the range of 400–670°C (Figs. 2A–2C). In many samples, where only a HTC is detected, its direction cannot be statistically distinguished from an ATC. The best degree of grouping of the ATC ($k_{max} = 74.5$) is reached at 75% of step-wise unfolding; i.e., it is synfolding. We know that the major deformations in the Western Urals occurred in the Permian [11], and the synfolding magnetization is likely to have formed precisely at that time. A part of the studied rocks was probably completely remagnetized. In such samples the Permian trend is manifested as an HTC. If remagnetization was partial, the Permian trend is detected as an ATC.

The comparison of the pole determined from the trend of synfolding magnetization with the apparent polar wander path for Baltica showed that the first pole coincides with the model poles for Baltica with ages of 270 and 280 million years (Fig. 3). This fact not only confirms the assumption that synfolding magnetiza-

Geological Institute (GIN), Russian Academy of Sciences, Pyzhevskii per. 7, Moscow, 109017, Russia
University of Florida, Gainesville, USA
Zavaritskii Institute of Geology and Geochemistry, Ural Branch, Russian Academy of Sciences, Yekaterinburg, Russia
e-mail: namife90@gmail.com



(C)

Upper Ediacaran Sylvitskaya series	Ust'-Sylvitskaya suite
	Chernokamenskaya suite 557 ± 13
	Perevalokskaya suite
	Staropechninskaya suite
Lower Ediacaran, Serebryanskaya series	Kernosskaya suite
	Butonskaya suite
	Koivinskaya suite
	Garevskaya suite
	Taninskaya suite

Fig. 1. Contour maps of Europe (A) and the Middle Urals (B) and the stratigraphic column for the sections of the Ediacaran deposits (C). The age, in millions of years, is indicated for the Chernokamenskaya suite. (*1*) is the Mesozoic–Cainozoic deposits of the Urals; (*2*) is the Paleozoic deposits of different tectonic zones in the western slope of the Middle Urals; (*3*) is the Paleozoic deposits of the eastern slope of the Urals; (*4*) is the Sylvitskaya series; (*5*) is the Serebryanskaya series; (*6*) is the metamorphosed pre-Ediacaran deposits; (*7*) is the Troitsk granitic syenitic rock mass; (*8*) is the tested intervals; (*9*) is the Middle Urals.

tion is acquired by the Chernokamenskaya suite rocks during the Permian period, but also allows us to state that the region under consideration was not subject to block rotations relative to Baltica in the Phanerozoic.

We managed to detect an HTC directed to the origin of coordinates and different from the trend of Permian remagnetization in approximately one-third of the collection that we studied (Figs. 2A–2C). The blocking temperatures reach 680°C in the majority samples, but sometimes are less than 580°C. Hematite, or the less frequently occurring magnetite, is obviously the main magnetization carrier in the studied rocks. The directions of the HTC are grouped quite well within the sites and the entire collection (Figs. 2D and 2E; Table 1). The fold test [12] is not informative, since the HTC trend is close to the strike of the studied rocks.

We identified the trends for the HTC of both direct and reverse polarities (Figs. 2D, 2E). For primary magnetization under sufficient averaging of the interstratum scatter, their trends should differ by approximately 180°. The available data on reverse polarity are insufficient to perform the test correctly. However, it looks positive, which is usually considered as a positive argument in favor of the primary character of the identified magnetization.

There are also other arguments for the above-mentioned. If we compare the data obtained with the apparent polar wander paths for Baltica [13, 14], we will see that the pole of the Chernokamenskaya suite is far from these paths in the direct and reverse polarity (Fig. 3). If the studied rocks were remagnetized in the Phanerozoic, the pole would have coincided with a respective segment of the path. By summing up the above information, we may consider with a sufficient degree of confidence that the component that was detected last in the rocks of the Chernokamenskaya suite is primary.

The western part of the Ural fold belt is a deformed marginal plate, and there were no significant displacements of the Kvarkushsko–Kamennogorksii anticlinorium relative to the plate. This is confirmed by the similar formations in the western part of the Ural fold belt with the deposits of the plate itself, as well as by the fact that the sedimentary rocks of the upper Riphean and Ediacaran in the western slope of the Middle Urals are reliably compared with the deposits outcropped by the deep boreholes in the Pre-Ural foredeep and the eastern section of Baltica [15]. As we

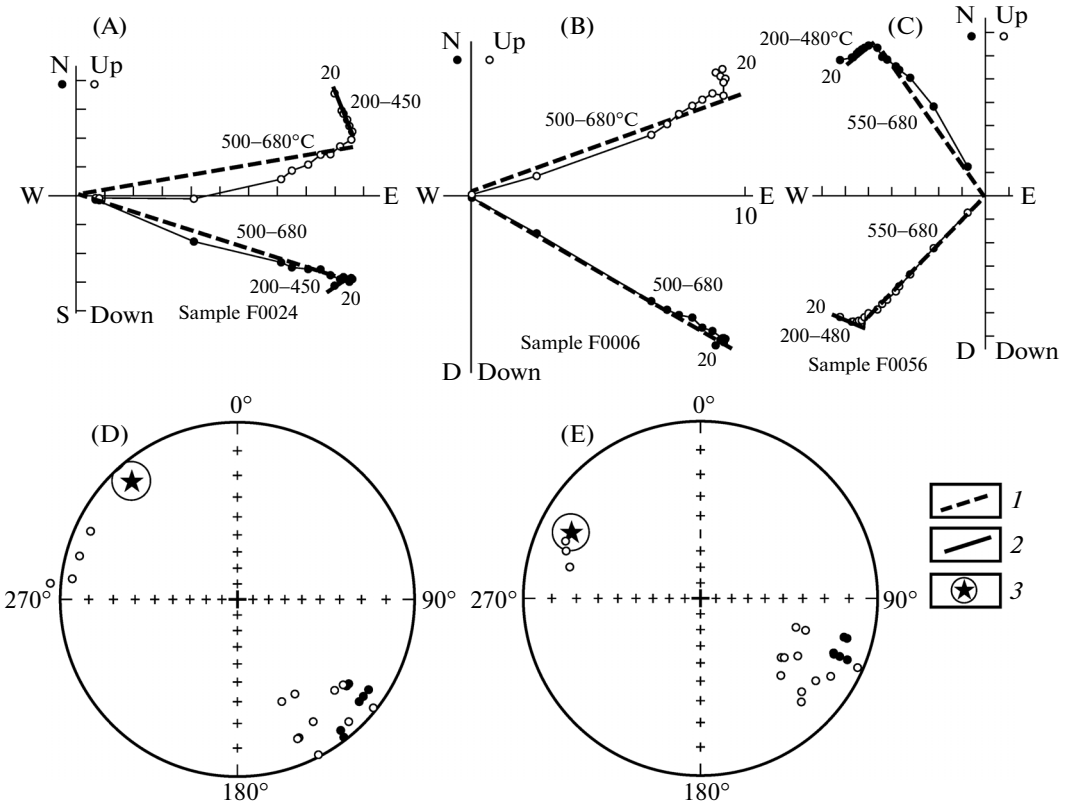


Fig. 2. (A—C) are the Zijderveld diagrams for the samples from the Chernokamenskaya suite in the stratigraphic coordinate system in the direct (A, B) and reverse (C) polarity. The shaded (unshaded) circles are the projection on the horizontal (vertical) plane. (D and E) are the stereograms of HTC magnetization directions for the rocks from the Chernokamenskaya suite in the geographic (D) and stratigraphic (E) coordinates. The shaded (unshaded) circles are the directions of the direct (reverse) polarity. (*1*) high-temperature component (HTC), (*2*) average temperature component (ATC); (*3*) mean direction of the HTC with a confidence circle.

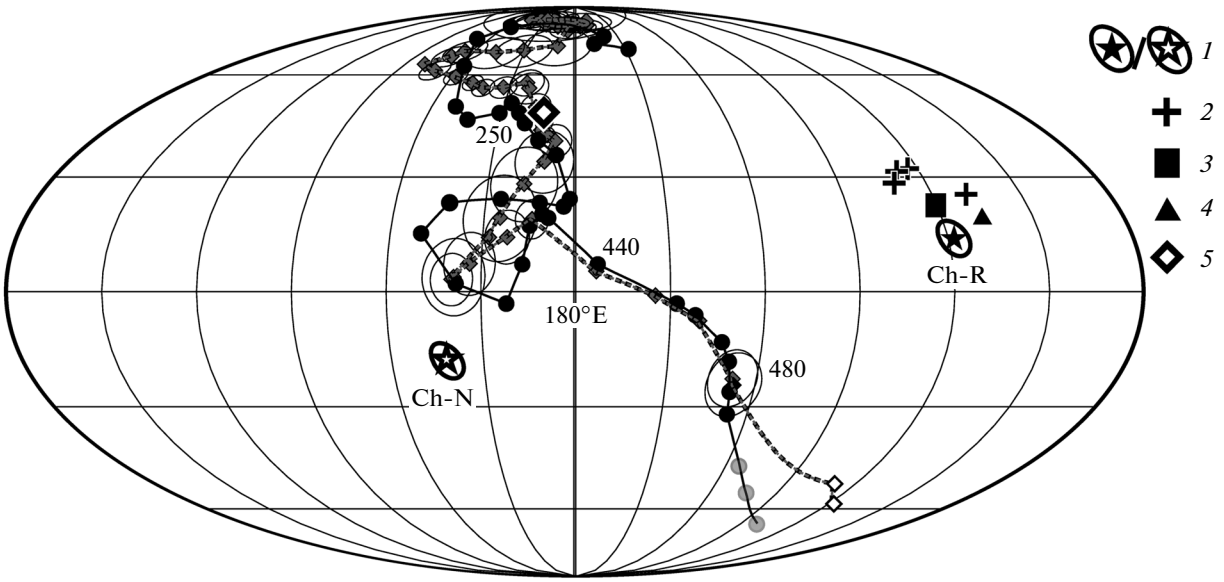


Fig. 3. The apparent polar wander paths for Baltica from the Early Ordovician till the present time and some Ediacaran data for this plate. The black solid line with circles designates the curve from [8]; the black dashed line with rhomboids denotes the one from [9]. (*1*) pole for the rocks from the Chernokamenskaya suite for the two variants of polarity selection (Ch-N and Ch-R, keys are given in the text); (*2*) poles aged approximately 555 million years, the southeastern coast of the White Sea [12–14]; (*3*) is the pole aged 540–550 million years for the volcanic rocks from the Southern Volyn' [15]; (*4*) is the pole aged 540–550 million years for the sedimentary rocks from the Southern Urals [11]; (*5*) is the pole for the secondary component in the rocks from the Chernokamenskaya suite.

Directions of the magnetization primary component

Site	DA, deg	DD, deg	N/No	MSC				ASC			
				D, deg	I, deg	k	α_{95}, deg	D, deg	I, deg	k	α_{95}, deg
F0001	126	72	7/7	152.5	−0.6	25.1	12.3	133.8	−25.7	25.1	12.3
F0008	126	72	7/6	148.2	−11.8	27.7	13.0	120.6	−24.9	27.7	13.0
F0015	126	72	7/7	133.1	−16.2	74.4	7.0	112.3	−11.8	74.4	7.0
F0022	126	72	7/6	137.7	−3.9	37.6	11.1	125.6	−12.7	37.6	11.1
F0029	126	72	7/5	156.0	−8.9	24.6	15.7	126.2	−31.1	24.6	15.7
F0036	127	72	6/3	142.3	1.0	79.4	13.9	132.7	−14.3	81.9	13.7
F0043	127	72	14/8	148.7	−26.0	25.8	11.1	106.9	−26.4	28.5	10.6
F0057	127	73	7/6	141.7	3.4	47.3	9.8	135.8	−11.6	52.4	9.3
F0064	128	76	7/4	156.5	−25.7	45.5	13.8	106.9	−31.2	33.0	16.2
F0068	128	77	6/3	156.7	−8.8	121.8	11.2	125.3	−29.8	91.8	12.9
F0132	125	34	7/6	280.4	−4.1	33.6	11.7	286.6	−17.4	35.5	11.4
F0139	127	33	6/5	284.4	−3.3	35.8	13.0	289.4	−15.0	37.7	12.6
F0145	124	34	6/5	290.4	−5.0	78.0	8.7	295.5	−11.8	79.4	12.6
N6205	299	88	8/8	128.4	13.6	29.8	10.3	105.1	10.0	29.8	10.3
N6213	299	88	7/7	130.0	6.2	45.1	9.1	112.7	11.6	45.1	9.1
N6221	299	88	7/7	127.1	13.5	24.8	12.4	105.3	8.8	24.8	12.4
N6228	299	88	7/5	124.5	6.1	143.5	6.4	112.7	6.1	143.5	6.4
N6235	120	76	8/5	128.6	−1.1	51.1	10.8	120.9	−8.8	39.5	12.3
N6243	129	84	9/7	128.9	−15.0	35.6	10.2	113.8	−1.8	40.9	9.5
Mean (sites)			19	314.5	3.3	15.9	8.7	297.2	8.0	16.8	8.4

Comment: MCS is the modern coordinate system; ACS is the ancient coordinate system; DA is the dip azimuth; DD is the degree of dip; N/No is the ratio of the selected samples to the used samples; D is the declination; I is the inclination; k is the concentration parameter; α_{95} is the radius of confidence circle.

showed, block rotations were not observed in the region either. Therefore, the Chernokamenskaya suite pole may be extrapolated over the entire plate.

We should note that the pole obtained is consistent with several same-age poles with respect to the sedimentary rocks of the Southern Urals [3] and the southeastern coast of the White Sea [4–6], as well as the volcanic rocks of the southern Volyn' [7]. These poles form a quite dense cluster by the regions within the boundaries of Baltica, which are remote from each other at a distance of 1500–500 km (Fig. 3). The mean pole calculated by this group has latitude P_{lat} = 25.5° N and longitude P_{long} = 300.3° E (concentration parameter K = 37.6, radius of confidence circle A_{95} = 10.0°, N = 7 poles). All these poles may be considered to be independent of other records of the Earth's magnet field, which means that the test (criterion) of the external convergence is positive and magnetization is primary in all studied objects.

Existence of this group of seven more or less even-aged poles permits claiming that at the end of Vendian the East European Platform was situated in subequatorial latitudes of the Northern or Southern hemispheres.

AKNOWLEDGMENTS

This study was supported in part by the Russian Foundation for Basic Research, project no. 11-05-00037, and NSF, project EAR11-19038, as well as Program no. 10, Department Earth Sciences, RAS.

REFERENCES

1. B. D. Ablizin, M. L. Klyuzhina, F. A. Kurbatskaya, and A. M. Kurbatskii, *Upper Riphean and Ediacaran of*

Western Slope of the Middle Urals (Nauka, Moscow, 1982) [in Russian].

2. D. V. Grazhdankin, A. V. Maslov, and M. T. Krupenin, Stratigr. Geol. Corell. **17** (5), 20–40 (2009).
3. N. M. Levashova, M. L. Bazhenov, J. G. Meert, N. B. Kuznetsov, I. V. Golovanova, K. N. Danukalov, and N. M. Fedorova, Precambrian Res. **236**, 16–30 (2013).
4. V. Popov, A. Iosifidi, A. Khramov, J. Tait, and V. Bachtadse, J. Geophys. Res. **107** (2002).
5. V. V. Popov, A. N. Khramov, and V. Bachtadse, Rus. J. Earth. Sci **7** (2), 1–29 (2005).
6. M. P. Iglesia Llanos, J. A. Tait, V. Popov, and A. Abalmasova, Earth Planet. Sci. Lett. **240**, 732–747 (2005).
7. S.-A. Elming, S. Kravchenko, P. Layer, O. Rusakov, A. Glevasskaya, N. Mikhailova, and V. Bachtadse, *J. Geol. Soc*. (London, 2007), Vol. 164.
8. Yu. L. Ronkin, D. V. Grazhdankin, A. V. Maslov, G. A. Mizens, D. I. Matukov, M. T. Krupenin, G. A. Petrov, O. P. Lepikhina, and A. Yu. Kornilova, Dokl. Akad. Nauk **411** (3), 354–359 (2006).
9. J. L. Kirschvink, Geophys. J. R. Astron. Soc. **62**, 699–718 (1980).
10. P. L. McFadden and M. W. McElhinny, Geophys. J. Intern **103**, 725–729 (1990).
11. V. N. Puchkov, Geol. Geofiz. **44** (1/2), 28–39 (2003).
12. M. W. McElhinny, Geophys. J. R. Astron. Soc. **8**, 338–340 (1964).
13. M. A. Smethurst, A. N. Khramov, and S. A. Pisarevsky, Geophys. J. Intern **133**, 44–56 (1998).
14. T. H. Torsvik and L. R. M. Cocks, Norw. J. Geol **85**, 73–86 (2005).
15. *The Riphean Stratotype. Stratigraphy. Geochronology*, Ed. by B.M. Keller and N.M. Chumakova (Nauka, Moscow, 1983) [in Russian].

Translated by L. Mukhortova